

Personal Computer World

MSX
Basic, the
new standard?

Canada \$2.75/US \$2.50/FF 18.50/FL 8.15/SFr 8.00/IR £1.23/BFr 99.00/
Lire 4,700/DKr 24.00/NKr 161.00/DM 9.5

March 1984 85p

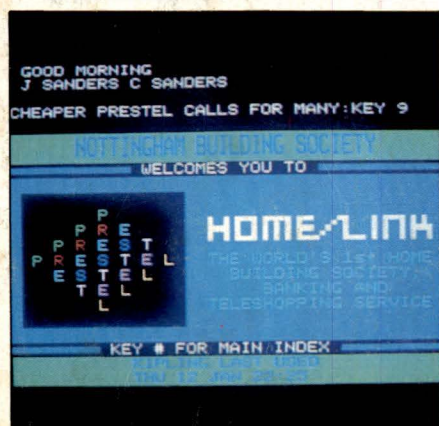
BRITAIN'S BIGGEST MICROCOMPUTER MAGAZINE



Inside Macintosh-Apple's latest



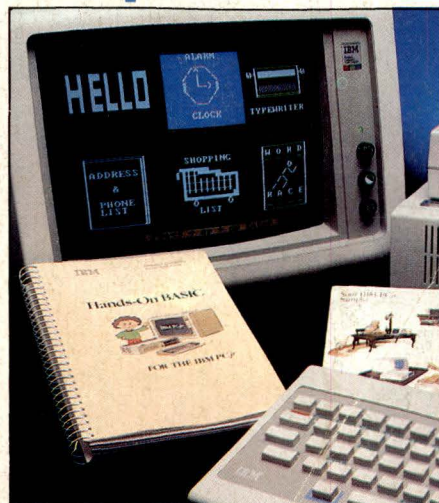
Improve your chances on the pools



Dial your bank



Compact Canon!



IBM's PC Junior

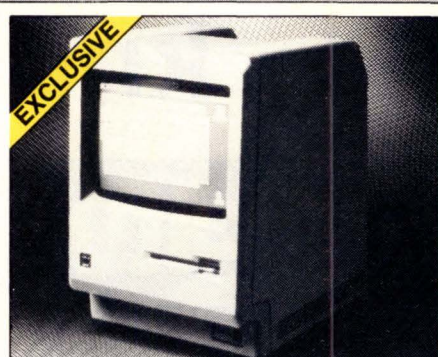
C • O • N • T • E • N • T • S



Canon X-07
A hand-held portable
Page 48



Spectravideo
MSX Basic in action
Page 72



Macintosh
Apple's promising comeback
Page 104

B E N C H T E S T S & R E V I E W S

CANON X-07	48	People are bound to buy it, but is it value for money?
New Japanese hand-held portable with infra-red communications.		
WHICH SPREADSHEET?	61	
PEACHCALC — A comprehensive package from a company that knows what it's about.		
SPECTRAVIDEO	72	
A competitor for the BBC Micro and Commodore 64 — this new machine features the aspiring standard MSX Basic.		
MACINTOSH	104	
Apple's bid to revitalise the company. This portable and powerful machine is part of the Lisa 'family' and employs user-friendly icons.		
IBM PC JUNIOR	168	
First full Benchtest this side of the Atlantic.		
		SCHOOLPLAY 224
		Pre-school packages and spelling tests for the BBC and Spectrum.
		HOMELINK 232
		The shape of things to come. A peek at the home of the future with online banking and shopping.
		STACK LIGHT PEN 240
		This light pen has business pretensions — could it steal the show from the mouse?
		PERSONAL DATA BASE 253
		An IBM PC package to get you going with data management.

F E A T U R E S

BRITISH MICROCOMPUTING AWARDS	29	
Full list of our and your nominations for the most important awards in the micro industry.		
POOLS FORECASTERS	96	
Can your micro get you rich quick? We test the success of the current offerings for home computers.		
TEACH YOURSELF ASSEMBLER	122	
This month looks at alternation as a building block in assembler programming.		
BRING A MICRO	176	
The machine takes on a new role in game playing. Get it to keep your score, and prompt you when you fall asleep between turns!		
UNIQUE UNIX	186	
Intro to the most widely-used, multi-user operating system for micros. Powerful but unfriendly.		
		MICRO ANIMATION COMPETITION 191
		Book reviews by John Melmoth and our challenging contest. There's still time to enter.
		LONG LIVE THE NEWBRAIN 200
		Focus on the NewBrain in business with hard disk, modem and serial to parallel interface convertor.
		APPLICATION PORTABILITY 204
		How one software house used the same code in its accounting package for a multitude of operating environments.
		THE AGE OF REASON 216
		The guise of future programming? Logo and Gesture in focus.
		PROFILE: HISTORY MAKER 223
		From the history of medicine to software marketing: one person's life story.
		A BEGINNER'S GUIDE TO PROGRAM CONVERSION 258
		Find out how to adapt everything to run on the Atari range.

BENCHTEST

In the spring of 1983, at the Hannover Messe (the enormous trade fair held in Northern Germany every year), Canon Incorporated demonstrated, next to its AS 100 desktop computer (Benchtested in the December 1983 issue of *PCW*), its brand new portable computer, the X-07. As happens so often in this industry, it has taken from then till now for this machine to reach the European market in any quantity. To paraphrase Harold Wilson, a week is a long time in the computer business, and a machine which eight months ago looked attractive, with some interesting and novel ideas incorporated, now looks much more ordinary. That said, the X-07 is still a machine which should find a place in the portable market, given what it offers for the price.

It is certainly an attractive machine to look at, with its silver and black body contrasting nicely with the white keys and their black symbols. With the dimensions of a thickish paperback book, the machine is possibly the smallest portable on the market. It comes in a sturdy, silver-coloured plastic carrying case which hinges open to allow operation and has cut-outs at the sides to give access to the various interfaces provided on the computer. The keyboard (Qwerty) is not full-size, nor are the individual keys, but it has a nice feel (albeit calculator-style). It does not, however, permit efficient touchtyping. I think that a slightly larger computer to make room for a proper keyboard, along the lines of the Epson HX-20 or the NEC PC-8201A, would have been a better choice.

However, its small size does mean that it can be slipped easily into a suit pocket or tucked away in the corner of a briefcase, so it should appeal to the travelling businessman wanting instant computing facilities.

Hardware

The machine measures 200mm × 130mm × 26.5mm and weighs only 480g without the four AA penlight batteries. On the left hand side is a serial interface, the back has a volume control and an expansion interface, and on the right hand side is a parallel port, a contrast control for the LCD screen, an AC adaptor socket and a cassette plug. Underneath there are three covered compartments: one to hold the four batteries; one to give access to the socket which receives the 8k extension RAM chip, and the third to hold the memory cards (of which more later).

A four line by twenty character LCD screen is mounted in the top left hand corner. Next to this is the grille for the tiny piezoelectric speaker and then what looks at first sight to be a stylised



Possibly the smallest micro on the market, the Canon X-07 runs on four AA penlight batteries.



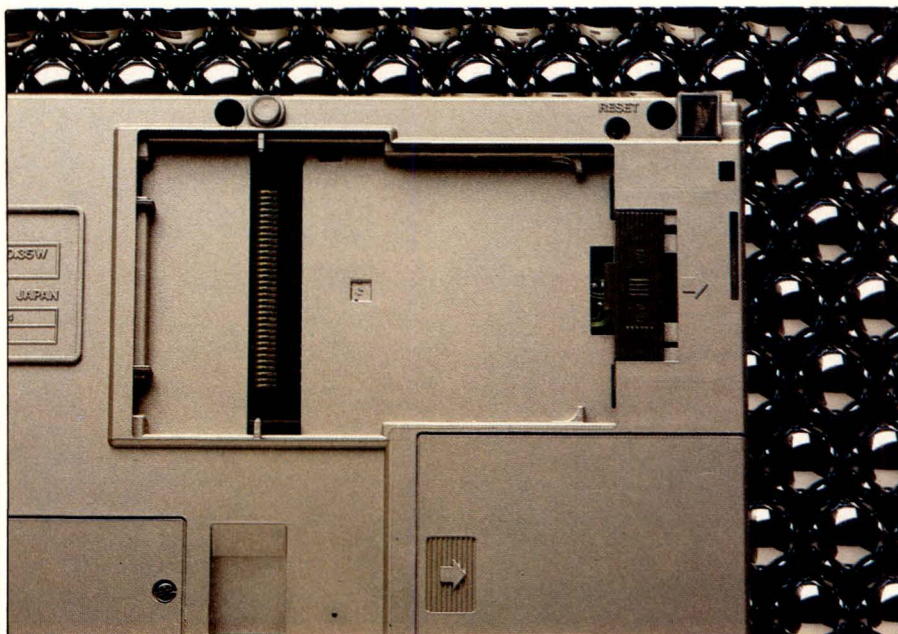
The 4-colour graphic printer/plotter plugs directly into the X-07.

flower logo but on closer examination proves to be a square white function key surrounded by the up, down, left, right cursor control keys. Then there are five blue function keys mounted under the screen and labelled F1 to F5; to the side of these are four white keys marked respectively INS(insert), DEL(delete), HOME/CLR(home cursor, clear screen) and OFF; and finally an orange coloured ON/BREAK key. Underneath this line of keys is the keyboard proper, in almost standard Qwerty layout. There is only one SHIFT key, which has been placed below the keyboard to the left of the relatively small space bar; two extra keys have been placed between the

SHIFT key and the space bar — NUM (numeric keypad) and GRPH(graphic characters); and lastly, to the extreme right of the space bar is the RETURN key, white on blue.

The basic machine comes with 8k of RAM, but the Basic steals some of this, so that when you switch it on you are told that you have 6761 bytes free. The RAM can be expanded to a maximum of 24k. Since all the RAM is battery-backed CMOS, whatever is in the memory is retained when the machine is switched off. The processor is the NSC800, the CMOS version of the Z80 and there is also a second processor, the T5834, a custom made CMOS chip for screen

BENCHTEST



Underneath the machine a recessed compartment allows the insertion of 'memory cards'.

and keyboard control. The Microsoft Basic interpreter is held in 20k of ROM, and the ROM area can be increased to 42k. 12k of memory has been set aside for use by an expansion box which will, among other things, drive a full-size screen.

A tiny piezoelectric speaker has been built into the machine but, mercifully, has a volume control, so that the beep which emanates from it every time you hit a key can be turned down low or even below the threshold of hearing. The sound can also be turned off by a Basic command, `CONSOLE`. The LCD screen displays four lines of twenty characters which now seem rather small in comparison to the NEC PC-8201A with its 40×8 format. Each character is formed in an eight high by six wide matrix, giving a pixel count of 120×32 .

Graphics are also possible on the display, with each point being individually addressable. A contrast control is also provided for the screen, so that it is possible to view it from different angles, which is a useful touch on a hand-held (or lap-placed) machine. The display scrolls vertically but not horizontally. This is a shame, since a window onto a larger screen (Osborne style) would have gone some way to offset the disadvantages of the small LCD. Six function keys (with the aid of the `SHIFT` key) make it possible to have twelve programmable functions available. Five of these keys are situated under the screen and the last line can be used to display the function of each key. These five keys contain default functions which can be changed. These are: `PRINT TIME$`,

`PRINT DATE$`, `CLOAD`, `CSAVE`, `LOCATE(X-Y` cursor control), `PRINT`, `LIST`, `SLEEP` (turns the machine off but retains contents of screen), `RUN` and `CONT(inue)`. The sixth function key, in the middle of the cursor control cluster, is empty. Use of the `NUM` and `GRPH` keys extends the keys available. The `NUM` toggles on and off a numeric keypad. This is formed by the original 7, 8 and 9 keys plus the U, I and O keys which become 4, 5 and 6, the J, K and L keys which become 1, 2 and 3, and the M key which becomes 0. I do not regard this as a facility that I would greatly miss, but perhaps there are people who have entered so much numeric data in their time that their fingers automatically stray to the right and this might be of some use to them.

The `GRPH` key brings into play a whole range of special characters, a generous sprinkling of the Greek alphabet, all the accented characters needed in French and German, the pound sign, the hearts, clubs, diamonds and spade symbols and one or two others. To show you the character on each key, Canon has provided an overlay which I have managed to lose twice already. It seems to me that it would have been a lot simpler to have printed the graphic characters at the side of each key directly on to the body of the computer. The effect would be exactly the same and would prevent a lot of swearing on the user's part!

The ASCII codes from 128 to 255 are used for special characters and two blocks of these — 128 to 159 and 224 to 255 — can be redefined by the user with the Basic function `FONT$`. The block in

the middle is reserved for Japanese characters.

Cassette

Saving to, and loading from, cassette proved to be no problem. I was using a small Tandy model and it quite happily cooperated at the 1200 baud transfer rate used by the X-07. The volume control was set around the half-way mark, but there was a fair margin either side at which it continued to work. The standard commands `CLOAD` and `CSAVE` are used, plus a verification command, `CLOAD?`. Given the small amount of storage (8k) available on the removable RAM cards, cassette tape must remain the best storage medium for the X-07, unless Canon intends to come out with some form of micro-drive.

Memory cards

One of the innovative features of the Canon X-07 is the facility to add to, or change part of, the memory by means of a memory card. This card, which has similar dimensions to a 'credit-card' calculator, and looks like one with the display and keyboard removed, is held in a compartment underneath the X-07. The cover of the compartment has a spring-loaded latch on which is mounted a power ON/OFF switch. Only



The X-721 optical coupler facilitates communications with other peripherals.

B E N C H T E S T

when this switch is in the OFF position is it possible to slide the latch to one side and remove the cover. The card is then simply placed in the recess, connector side down, the cover replaced and the power switch set to ON. There are four different types of cards available: 4k and 8k CMOS RAM cards, each containing a lithium battery for back-up when the card is removed from the computer; 8k ROM cards; and 8k ROM/4k CMOS RAM cards, also lithium battery-backed. The ROM and ROM/RAM cards are used to provide pre-written programs and applications.

It is possible to set aside part of the RAM of the X-07 to hold files. Logically enough, this is called a RAM FILE area in the manual. Using the Basic command FSET, parameter, one can reserve any amount of RAM to this purpose up to the total RAM available less 87 bytes. This area is always taken from the top of RAM downwards. A 4k or 8k RAM card will always sit on top of the built-in RAM, and be contiguous with it, so the FSET command can be used to ensure that all the files you create will be held in the RAM card. For example, with a 4k RAM card in place, the command FSET 4096 will turn the card into a RAM FILE area. SAVE filename will place any program you create in the card, which can then be removed from the machine and stored safely for up to eighteen months, protected by the lithium battery-backup. Just as in CP/M, the DIR command will list all the files in the RAM FILE area. The programs can be run by typing LOAD "filename" and then RUN, or simply RUN "filename".

Memory map

FFFF	Basic ROM
B000	For Extension
8000	Optional Memory Card (ROM)
6000	Optional Memory Card (Memory Socket)
4000	Optional Memory Card Or Socket (RAM)
2000	
0000	RAM

Peripherals

One of the features that is still unique to the X-07, as far as I know, is the capability to communicate with peripherals or, indeed, with other X-07s, without physical connection. This is accomplished using the X-721 optical coupler, which plugs into the serial interface at the side of the X-07 and provides infra-red communication with a similarly equipped X-07 or, via a second X-721 and an RS232C level convertor (X-722), with a peripheral device, such as a printer. Transmission speed can be set between 100 and 2400 bauds and the maximum range is approximately five metres. Unfortunately, this device was not available at the time of the Benchtest, but I did see one in action at the distributors linking an X-07 to a printer and it did appear to be performing faultlessly at a distance of about three metres. (See the photos here.)

Two printers are available for plugging directly to the X-07. There is a 4-colour graphic printer/plotter, with specifications remarkably similar to that of the Oric MCP-40: prints in red, green, black and blue; 114mm wide plain paper, up to eighty characters per line; X-Y plotting; a printing speed of ten characters per second; and a Centronics interface. There is also a small thermal printer which prints up to 35 characters per line at a speed of 20 characters per second onto 57mm wide paper through the serial interface.

I mentioned earlier the RS232C level converter, model X-722. This is used to provide a standard 25-pin serial interface for connection to various peripherals. It is needed since the serial output from the X-07 has only the minimum nine pins needed to conform to RS232C specifications. It also provides power to the optical coupler used at the peripheral end of an infra-red link.

Finally, there will be an expansion box principally to provide output to a full-size monitor screen, but this is not yet available.

Basic

The version of Microsoft Basic implemented on this machine lacks one or two things that I miss, like AUTO and RENUM, but it does have a screen editor, even if it is a little screen. It also has a few additions which are worth describing:

TIMES and DATES. These take advantage of the built-in clock.

STARTS. This is an auto-start string, and is best demonstrated by giving an example of its use. By putting the commands RUN and RETURN in the STARTS and then turning the machine

off not with the normal OFF button but with an OFF1 command, any program left in the text area of RAM will be run whenever the machine is turned on. This function can be suspended with the OFF2 command.

ALMS. This is an alarm which can be set for the year, month, day, day of the week, hours and minutes. It will sound for one minute and functions even if the computer is switched off. It is set by one of the parameters of the CONSOLE command.

SCREEN (X, Y). This returns the ASCII code of the character located at the position indicated by the X, Y parameters.

STICK. This returns the status of the cursor keys. 0 will be returned if nothing is depressed, 1 if the UP key is depressed and 2 if the UP key and the RIGHT key are depressed simultaneously, 3 if the RIGHT is depressed and so on around the clock up to a value of 8.

LOCATE. This positions the cursor at one of the eighty character positions on the screen.

CONSOLE. This has several parameters allowing you to: set the line at which you wish scrolling to start and the number of lines you want to scroll; display the labels of the function keys below the keyboard; activate or silence the beep which accompanies a key press; and activate or cancel the key auto-repeat.

CONSOLE@. This has parameters to turn the alarm on and off and to set the keyboard to the normal alphanumeric mode, Kana mode (Japanese characters), graphic mode or ten-key mode.

STRIG. This is an unusual one. It returns the status of the space bar and function key 6 (the one in the middle of the cursor control cluster). A = STRIG(0) will put -1 in A if the space bar is depressed, 0 if not. A = STRIG(1) works similarly for function key 6. I assume it is to be used in conjunction with STICK for games programs.

Finally, there are several graphic commands and one sound command:

CIRCLE. That's right, it draws a circle.

LINE. You've got it!

PSET. Puts a dot at a specified X-Y coordinate.

PRESET. Erases same.

POINT. Checks for a dot.

BEEP. This is the sound command. It has only two parameters, tone and duration and covers four octaves. Despite its single-note limitations and restricted range, my wife had a lot of fun entering and running a program from the manual which turned part of the computer keyboard into a piano keyboard.

As the Benchmarks show, the Canon

B E N C H T E S T

version of Microsoft Basic is not particularly sparkling, being considerably slower than the Hewlett Packard HP-75C, the Epson HX-20 and the Tandy Model 100, and being positively put to shame by the results obtained by David Tebbutt with the NEC PC-8201A.

This is even more surprising when one considers that the clock rate of the X-07 is 3.84MHz. This is, I feel, a serious drawback to this machine, since the type of customer who is prepared to pay out the money to have computing capacity at his/her fingertips is also likely to be the person who wants quick results from said computer, and a Basic this slow is not going to give it.

Documentation

The documentation takes up more room than the box in which the computer was delivered. There are three glossy-covered A5-size manuals: *Programming For Beginners*, *User's Guide* and *Basic Reference Manual*. There is also a 16-page folding card for quick reference to the Basic commands and functions. *Programming For Beginners* contains 210 or so pages, clearly laid out in two colour (light blue and black) text, diagrams and 'amusing' sketches. Chapter one explains what a program is, encourages the use of flowcharts, gives a few tips on writing efficient programs, and shows how to save in the RAM file area and on cassette. Chapter two runs through basic Basic. Chapter three describes several practical programs to consolidate what was learned in Chapter two, while Chapter four contains programs which demonstrate the different features of the X-07: sound; graphics; colour printer; optical coupler, and so on. Chapter five is dedicated to a games program of interplanetary commerce called, appropriately enough, 'Space Trader'. It provides a listing of the complete program and also explains in detail how the program works.

The manual is clearly written, in good, if not absolutely perfect English, and is much better than some Japanese-written 'English' manuals I have read.

The User's Guide, which is the thinnest offering at 136 pages, starts with the unpacking of the machine and then goes on to explain its functions. It also contains details on the internal workings of the Basic, listing things such as the addresses for each pointer and the pattern and format of the variable area. At the very back of the manual can be found all the interface specifications, which include the pin-outs for the serial, parallel, cassette and expansion interfaces.

The Basic Reference Manual, 180-

odd pages long, is fairly standard issue, with a couple of chapters dedicated to the 'grammar' of Basic, then a description of the commands in alphabetical order, followed by the functions. A useful index, which groups instructions and functions under headings such as 'Graphic Instructions', 'Variable Related Instructions' and 'File Related Instructions' is also included.

Overall, I think that the documentation is more than adequate and a welcome relief to someone who has suffered the rigours of CP/M manuals.

Applications software

There are three application programs which will be available initially for the X-07, all of them on the drop-in memory cards: File Card — a file management program; Table Card — for the preparation of table layouts; and Graph Card — for the graphic presentation of data, pie charts, bar charts, line graphs, and so on.

Prices

Canon X-07 hand-held computer (8k)	£ 265
8k memory extension (CMOS RAM)	80
4k memory card (CMOS RAM)	45
8k memory card (CMOS RAM)	85
8k memory card (RAM)	47
12k memory card (8k ROM/4k CMOS RAM)	N/A
X-721 optical coupler (infra-red)	86
X-722 RS232C level convertor	53
X-7104-colour graphic printer	170
X-711 thermal printer	82
XP-110 File card	48
XP-120 Table card	48
XP-130 Graph card	48
8k memory extension (CMOS RAM)	45

The X-07 is not yet available in the UK. Sterling prices given here are approximate and have been converted from Dutch guilders.

Conclusions

I enjoyed using the machine, and particularly appreciated its portability. However, this portability is severely restricted by the life of the four AA batteries, which only allow about eight hours of operation. (With the machine switched off, they provide a back-up capability for the CMOS RAM of about one thousand hours.) An AC adaptor is available, but then the machine is no longer truly portable.

The RAM cards are a good idea, but their limited capacity prevent them from being a viable alternative to other storage media — that is, tapes and disks.

The infra-red communication capability is also a good idea, but its usefulness would seem to be limited to a medium-size office where you need to switch frequently between two different printers — say, a daisywheel and a dot-matrix, otherwise a cable would do just as well. The size of the machine makes it very convenient to slip into a briefcase but prevents it from taking a full-size keyboard. This precludes its use as a word processor, which is another application that Canon is claiming for it.

Canon talks about one of the applications of the X-07 being its use as an 'electronic notepad'. This, I think, is the best description of it. If this idea is pushed then, with Canon's well-known name in the industry, it should sell, but will hardly set the world alight. **END**

Benchmarks

BM1	3.9
BM2	11.5
BM3	32.0
BM4	35.8
BM5	38.1
BM6	58.3
BM7	78.9
BM8	379.6

All timings in seconds. For a listing of the Benchmark programs see 'Direct Access'.

Technical specifications

CPU	NSC800 (CMOS Z80)
RAM	8k (expandable to 24k)
ROM	20k (expandable to 42k)
Display	Four lines of 20 characters, 32 × 120 pixels
Keyboard	68 keys, including six function keys (giving 12 functions with SHIFT key). Graphic character access via GRPH key
External storage	RAM (interchangeable lithium battery-backed cards). Cassette (600 baud)
I/O	Centronics parallel, RS232, Cassette, Expansion
Built-in software	Microsoft Basic
Applications	File Card (Database); Table Card (Spreadsheet); and Graph Card (Graphics)
Dimensions	200mm × 130mm × 26.5mm
Weight	480g (excluding batteries)
Power supply	Four AA penlight batteries or AC adaptor